THE POLARIZED STACK IS FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition categories-definition-fibred-groupoids (canonical tag 003T). cited at proof line 4044. The conditions being checked.
- Lemma morphisms-lemma-ample-base-change (canonical tag 0893). cited at proof line 4054. Ampleness of the pullback.
- Lemma categories-lemma-fibred-over-fibred (canonical tag 09WV). cited at proof line 4076. Composition of fibred functors.
- Lemma categories-lemma-fibred-groupoids (canonical tag 003V). cited at proof line 4080. Fibre groupoid check.

Source reference. Original definition 003T: sources/stacks-chow/tags.txt, source lines 147–147. Cited at proof line 4044: the two conditions checked.

Source reference. Original lemma 0893: sources/stacks-chow/tags.txt, source lines 10483–10483. Cited at proof line 4054: L' is ample on X'/S'.

Source reference. Original lemma 09WV: sources/stacks-chow/tags.txt, source lines 12508–12508. Cited at proof line 4076: fibration over Sch_fppf.

Source reference. Original lemma 003V: sources/stacks-chow/tags.txt, source lines 149–149. Cited at proof line 4080: fibre categories are groupoids.

Lemma 0.2. *The category $\mathcal{P}olarized$ is fibred in groupoids over $\mathcal{S}paces'_{fp,flat,proper}$. The category $\mathcal{P}olarized$ is fibred in groupoids over Sch_{fppf}.*

Proof. We check conditions (1) and (2) of Categories, Definition 003T.

Condition (1). Let $(X \to S, \mathcal{L})$ be an object of $\mathcal{P}olarized$ and let $(X' \to S') \to (X \to S)$ be a morphism of $\mathcal{S}paces'_{fp,flat,proper}$. Then we let $\mathcal{L}'$ be the pullback of $\mathcal{L}$ to X'. Observe that X, S, S' are schemes, hence X' is a scheme as well (as the fibre product of schemes). Then $\mathcal{L}'$ is ample on X'/S' by Morphisms, Lemma 0893. In this way we obtain a morphism $(X' \to S', \mathcal{L}') \to (X \to S, \mathcal{L})$ lying over $(X' \to S') \to (X \to S)$.

Condition (2). Consider morphisms $(f, g, \varphi) : (X' \to S', \mathcal{L}') \to (X \to S, \mathcal{L})$ and $(a, b, \psi) : (Y \to T, \mathcal{N}) \to (X \to S, \mathcal{L})$ of $\mathcal{P}olarized$. Given a morphism $(k, h) : (Y \to T) \to (X' \to S')$ of $\mathcal{S}paces'_{fp,flat,proper}$ with $(f, g) \circ (k, h) = (a, b)$ we have to show

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there is a unique morphism $(k, h, \chi) : (Y \to T, \mathcal{N}) \to (X' \to S', \mathcal{L}')$ of $\mathcal{P}olarized$ such that $(f, g, \varphi) \circ (k, h, \chi) = (a, b, \psi)$. We can just take

$$\chi = \psi \circ (k^*\varphi)^{-1}$$

This proves condition (2). A composition of functors defining fibred categories defines a fibred category, see Categories, Lemma 09WV. This we see that $\mathcal{P}olarized$ is fibred in groupoids over Sch_{fppf} (strictly speaking we should check the fibre categories are groupoids and apply Categories, Lemma 003V). $\square$